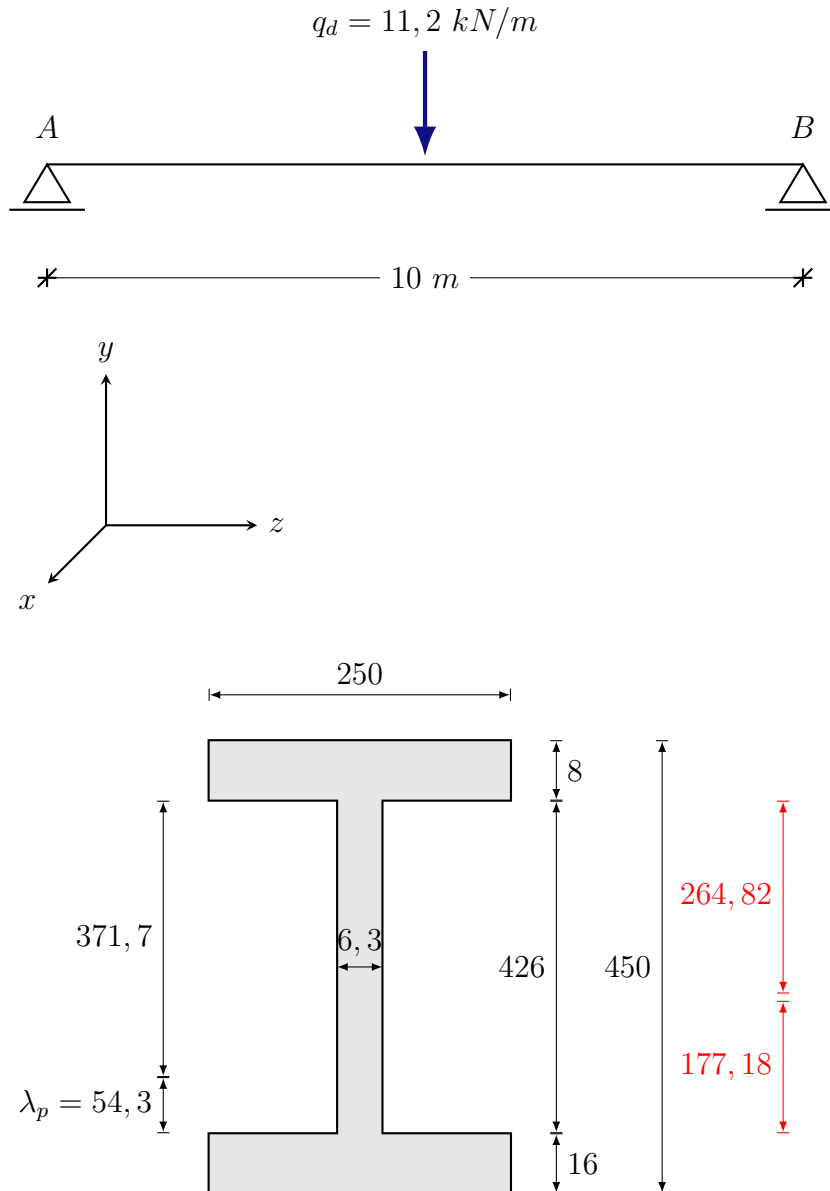


Exercícios

1. Verificar FLA , FLM , FLT para a viga birrotulada VSM 450 x 68, aço USICIVIL 350. Considere já os valores já majorados.



Dimensões em mm .

$$A_g = 86,8 \text{ cm}^2$$

$$I_y = 31,26 \text{ cm}^4$$

$$r_y = 6 \text{ cm}$$

$$I_x = 30691 \text{ cm}^4$$

$$C_w = 1332250 \text{ cm}^6$$

$$W_{xs} = W_{xc} = 1125 \text{ cm}^3$$

$$W_{xi} = W_{xc} = 1732 \text{ cm}^3$$

$$Z = 1445 \text{ cm}^3$$

$$J = 42 \text{ cm}^4$$

FLA - tabela D1 p. 144

$$\lambda = \frac{h_c}{t_w} = \frac{2 \times 264,82}{6,3} = 84,06$$

$$\lambda_p = \frac{\frac{h_c}{h_p} \sqrt{\frac{E}{f_y}}}{(0,54 \frac{M_{pl}}{M_r} - 0,09)^2} \leq \lambda_r$$

$$\lambda_p = \frac{\frac{52,96}{74,34} \sqrt{\frac{20000}{35}}}{(0,54 \frac{50575}{39375} - 0,09)^2} \leq \lambda_r$$

$$\lambda_r = 46,74 \leq \lambda_r$$

$$\lambda_r = 5,70 \sqrt{\frac{E}{f_y}} = 5,70 \sqrt{\frac{20000}{35}} = 136,25$$

$$y_{CG} = \frac{\sum Ay}{\sum A}$$

$$A_1 = 8 \times 250 = 2000 \text{ mm}^2$$

$$A_2 = 6,3 \times 426 = 2683,8 \text{ mm}^2$$

$$A_3 = 16 \times 250 = 4000 \text{ mm}^2$$

$$y_1 = 16 + 426 + 8/2 = 445 \text{ mm}$$

$$y_2 = 16 + 426/2 = 229 \text{ mm}$$

$$y_3 = 16/2 = 8$$

$$y_{CG} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3} = 177,179 \text{ mm}$$

$$h_p = 2 \times 37,17 = 74,34 \text{ cm}$$

$$M_{pl} = Z \times f_y = 1445 \times 35 = 50575 \text{ kNcm}$$

$$M_r = W \times f_y = 1125 \times 35 = 39375 \text{ kNcm}$$

Compacto λ

Semi λ_p

Esbelto λ_r

D 2.2

$$M_{Rd} = \frac{1}{\gamma_{a1}} \left(M_{pl} - (M_{pl} - M_r) \frac{\lambda - \lambda_p}{\lambda_r - \lambda_p} \right) = 41732,09 \text{ kNcm}$$

$$M_{Sd} = \frac{0,112 \times 1000^2}{8} = 14000 \text{ kNcm}$$

FLA OK

$$M_{Sd} \leq M_{Rd} \rightarrow 14000 \leq 41732$$

FLM - tabela D.1 p. 144

$$\lambda = \frac{b}{t} = \frac{\frac{250}{2}}{8} = 15,62 \text{ ver D.2.8-h}$$

$$\lambda_p = 0,38 \sqrt{\frac{E}{f_y}} = 0,38 \sqrt{\frac{20000}{35}} = 9,08$$

D2.8f - perfis soldados

$$\lambda_r = 0,95 \sqrt{\frac{E \times K_c}{70\% f_y}} = 0,95 \sqrt{\frac{20000 \times 0,48}{0,7 \times 35}} = 18,8$$

$$k_c = \frac{4}{\sqrt{\frac{h}{t_w}}} = \frac{4}{\sqrt{\frac{4,26}{0,63}}} = 0,48$$

$$\lambda_p \leq \lambda < \lambda_r$$

$$M_{Rd} = \frac{1}{1,1} \left(50575 - (50575 - 39375) \frac{15,62 - 9,08}{18,8 - 9,08} \right) = 39126,54 \text{ kNcm}$$

FLM OK

$$M_{Sd} \leq M_{Rd}$$

FLT tabela D.1

$$\lambda = \frac{L_b}{r_{ytc}} = \frac{1000}{4,9} = 204,08$$

$$\lambda_p = 1,76 \sqrt{\frac{E}{f_y}} = 42,07$$

D.2.8b

$$\lambda_r = \frac{1,38C_b\sqrt{I_yJ}}{r_{ytc}JB_1} \sqrt{B_2 + \sqrt{B_2^2 + \frac{27C_wB_1^2}{C_b^2I_y}}}$$

$$I_{Tcy} = \frac{(0,8 \times 25^3) + (37,17 \times 0,63^3)}{12} = 1042,44 \text{ cm}^4$$

$$A_{gt} = (8 \times 25) + (63 \times 37,17) = 43,4171$$

$$r_{ytc} = \sqrt{\frac{I_{Tcy}}{A_g}} = 4,9 \text{ cm}$$

$$C_b = \frac{12,5 M_{mx}}{2,5M_{mx} + 3M_A + 4M_B + 3M_C} R_m$$

$$M_{max} = M_B = \frac{ql^2}{8} = \frac{11,2 \text{ kN/m} \times 10^2}{8} = 14000 \text{ kNcm}$$

$$M_A = M_C = \frac{3ql^2}{32} = \frac{qlx}{2} - \frac{qx^2}{2} = \frac{3 \times 0,112 \times 1000^2}{32} = 10500 \text{ kN}$$

$$R_m = 0,5 + 2\left(\frac{I_{ym}}{I_y}\right)^2 = 0,5 + 2\left(\frac{\frac{8,25^3}{12}}{3126}\right)^2 = 0,72$$

$$C_b = 0,82$$

$$B_1 = \frac{(0,7f_y)w_c}{EJ} \leq \frac{f_yw_c}{EJ} \rightarrow B_1 = 0,0328 \leq 0,0072$$

$$B_3 = 0,45\left(d - \frac{t_{fs} + t_{fi}}{2}\right)(2\alpha_y - 1) = 0,45\left(45 - \frac{0,8 + 1,6}{2}\right)(2 \times 0,33 - 1) = -6,70$$

$$\alpha_y = \frac{I_{ytc}}{I_y} = 0,33$$

$$\beta_2 = \frac{5,2\beta_1\beta_3}{C_b} + 1 = -0,3938$$

$$\lambda_r = 119,63$$

$$M_{cr} = 11833,82$$

$$M_{Rd} = \frac{1}{\gamma_{a1}} M_{cr} = 10758 \text{ kNcm}$$

2. Verificar a viga em balanço, mesma viga do exercício anterior.

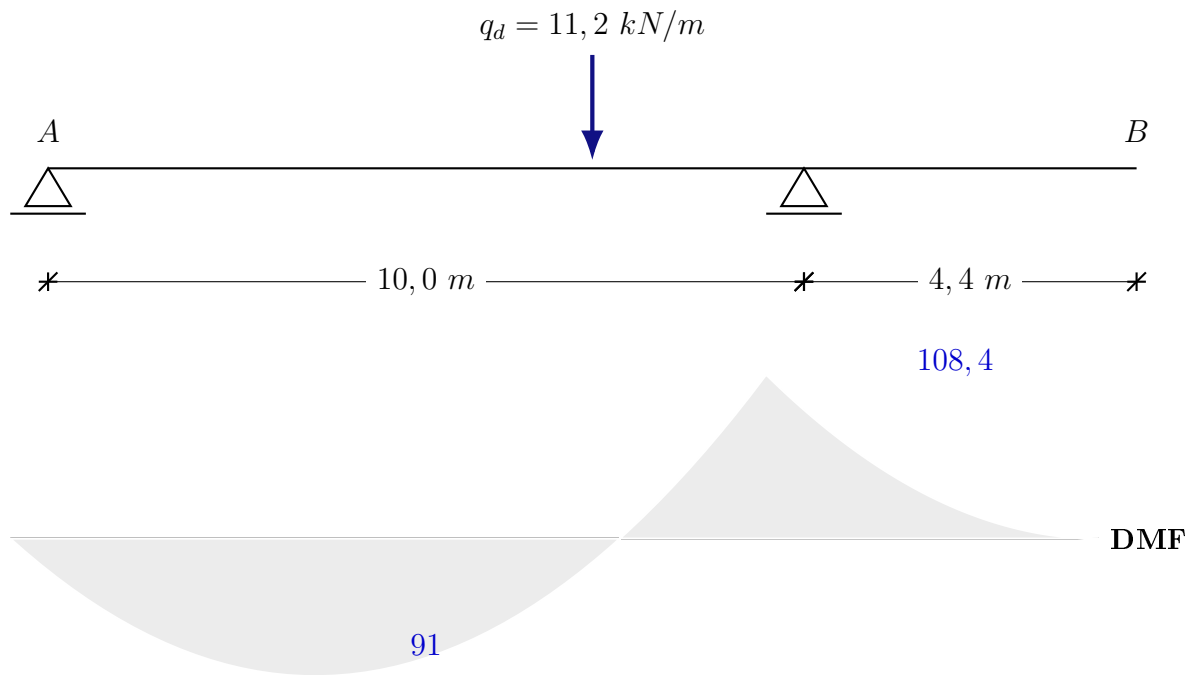
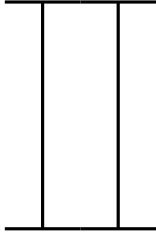


Diagrama de Momento Fletor.

Verificar FLA, FLM, e FLT.

3. Qual é o máximo momento fletor e cortante que uma viga de 20 m suporta? Com perfis justapostos. Carga? Considere carga pontual; 2 (VS 500 x 61).



FLM

$$\lambda \leq \lambda_p$$

$$M_{Rd} = \frac{M_{pl}}{\gamma_1}$$

$$\lambda = \frac{b}{t}$$

$$\lambda = \frac{\frac{b}{2}}{t}$$

FLA

$$\lambda_p \leq \lambda \leq \lambda_r$$